

ILE ROUGE, QC, Canada

WMO: 714260

Lat: 48.069N

Lon: 69.556W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.0	(c) -18.6	(d) -23.9	(e) 0.4	(f) -20.5	(g) -21.8	(h) 0.5	(i) -18.0	(j) 21.9	(k) -13.8	(l) 20.3	(m) -12.3	(n) 11.1	(o) 290	(p) 0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.5	(c) 18.1	(d) 15.0	(e) 17.0	(f) 14.3	(g) 16.0	(h) 13.6	(i) 15.9	(j) 17.3	(k) 15.0	(l) 16.4	(m) 14.1	(n) 15.6	(o) 5.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	16.6	14.2	10.1	15.7	13.3	9.6	14.9	44.5	17.4	41.9	16.5	39.7	15.6	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	15.7	13.7	-23.6	22.0	1.9	1.9	-25.0	23.4	-26.1	24.5	-27.2	25.5	-28.6	26.9
			-24.3	18.5	1.9	1.6	-25.7	19.7	-26.8	20.6	-27.8	21.6	-29.2	22.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-9.0	-8.0	-3.3	2.0	6.9	10.8	12.8	12.6	9.8	6.1	1.1	-4.9
	DBStd	8.35	5.67	5.16	4.46	2.83	2.22	2.16	1.94	1.89	2.17	2.41	3.59	4.82
	HDD10.0	2747	589	503	411	240	101	16	2	2	29	124	267	462
	HDD18.3	5548	848	737	669	490	356	226	171	178	255	381	517	720
	CDD10.0	239	0	0	0	0	3	39	89	82	23	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	6.6	8.8	8.6	7.8	6.5	5.4	5.2	4.9	4.9	5.5	6.4	7.3	8.3
(15) Precipitation	PrecAvg	1030	68	67	67	82	84	92	113	107	99	105	81	86
	PrecMax	1246	123	112	126	194	157	185	268	200	261	222	127	242
	PrecMin	805	5	36	31	14	30	36	53	48	51	33	44	26
	PrecStd	125	30	20	26	41	33	38	55	49	47	46	22	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.0	6.5	9.6	15.1	18.7	20.1	20.0	17.4	13.5	11.4	5.1
		MCWB	3.1	1.9	4.7	5.5	10.7	14.2	16.3	15.3	14.5	10.7	10.3	3.8
	2%	DB	1.7	1.4	3.7	7.5	13.1	16.7	18.2	17.9	15.5	11.8	7.8	3.2
		MCWB	1.0	0.5	1.9	4.7	9.6	13.4	15.4	14.8	13.4	9.8	6.6	2.2
	5%	DB	0.4	0.2	2.5	6.2	11.5	15.4	17.0	16.7	14.2	10.6	6.4	2.3
		MCWB	-0.3	-0.5	1.1	3.9	8.7	12.6	14.6	14.1	12.3	8.9	5.0	1.4
	10%	DB	-1.3	-0.7	1.5	5.2	10.2	14.2	16.0	15.7	13.1	9.5	5.3	1.3
		MCWB	-2.1	-1.4	0.4	3.2	7.8	11.7	13.8	13.5	11.4	7.9	4.0	0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.3	3.3	6.3	11.8	15.9	17.6	17.0	15.5	11.6	10.6	4.0
		MCDB	3.5	2.8	4.3	8.1	13.5	17.3	19.2	18.7	16.5	12.7	11.6	4.7
	2%	WB	1.2	0.8	1.9	5.1	10.1	14.5	16.1	15.6	13.7	10.2	7.1	2.4
		MCDB	1.7	1.3	2.7	6.7	11.9	16.0	17.5	17.2	14.9	11.4	7.8	3.0
	5%	WB	-0.2	-0.4	1.1	4.2	8.9	13.2	15.2	14.7	12.7	9.2	5.4	1.5
		MCDB	0.3	0.2	1.7	5.8	10.7	14.7	16.6	16.2	13.8	10.5	6.2	2.1
	10%	WB	-1.9	-1.5	0.4	3.5	7.9	12.2	14.3	13.8	11.6	8.2	4.2	0.5
		MCDB	-1.2	-0.9	1.1	4.9	9.6	13.8	15.8	15.4	12.8	9.5	5.1	1.1
(35) Mean Daily Temperature Range	MDBR		5.9	5.5	4.7	4.2	4.9	6.0	6.5	6.4	5.1	3.5	3.5	4.6
	5% DB	MCDBR	6.9	4.8	4.8	5.7	7.3	7.9	7.9	7.9	6.6	5.2	4.3	4.9
		MCWBR	7.1	4.7	4.1	4.0	4.9	5.5	5.6	5.6	5.0	4.1	4.3	4.4
	5% WB	MCDBR	7.0	4.8	4.2	5.2	7.0	7.7	7.5	7.6	6.4	5.2	4.4	4.5
		MCWBR	7.1	4.9	4.0	4.0	5.0	6.0	5.9	5.9	5.3	4.4	4.4	4.5
(40) Clear-Sky Solar Irradiance	taub		0.270	0.280	0.318	0.360	0.369	0.387	0.414	0.394	0.370	0.334	0.299	0.276
	taud		2.326	2.288	2.247	2.283	2.345	2.360	2.310	2.370	2.426	2.504	2.460	2.352
	Ebn at Noon		809	882	896	889	893	876	844	847	834	815	773	751
	Edh at Noon		70	92	114	122	120	119	123	111	96	76	63	61
(44) All-Sky Solar Radiation	RadAvg		1.29	2.21	3.39	4.42	5.11	5.48	5.42	4.68	3.61	2.11	1.15	0.87
	RadStd		0.15	0.26	0.25	0.37	0.50	0.38	0.53	0.31	0.25	0.18	0.12	0.08

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page